Appendix A
Hardness Measures Definitions

All measures are normalized to $[0, 1]$ range where applicable; Higher values generally indicate greater instance hardness unless otherwise noted

TABLE I: Instance hardness measures for classification problems

Name (Abbreviation)	Definition	Formula
Feature-based Measures		
Maximum Fisher's Discriminant Ratio (F1)	Measures the overlap between feature values in different classes by taking the largest discriminant ratio among all features. Higher values indicate better class separation.	$F1 = \frac{1}{1+\max_j r_j}$
Volume of Overlapping Region (F2)	Calculates the overlap of feature value distributions within classes. The product of normalized overlapping intervals across all features.	$F2 = \prod_{j=1}^{m} \frac{\text{overlap}(f_j)}{\text{range}(f_j)}$
Maximum Individual Feature Efficiency (F3)	Estimates the individual efficiency of each feature in separating classes, considering the maximum discriminative power among all features.	$F3 = \min_j \frac{n_o(f_j)}{n}$
Collective Feature Efficiency (F4)	Iteratively selects features with highest discriminative power between classes, measuring collective separability.	$F4 = \frac{n_o(f_{\max}(T_i))}{n}$
Neighborhood-based Measures		
k-Disagreeing Neighbors (kDN)	Proportion of k nearest neighbors with different class labels than the instance. Higher values indicate harder classification.	$kDN(x_i) = \frac{\lvert\{x_j \in kNN(x_i) \wedge y_j \neq y_i\}\rvert}{k}$
Fraction of Borderline Points (N1)	Percentage of vertices in a Minimum Spanning Tree that are incident to edges connecting examples of opposite classes.	$N1 = \frac{1}{n}\sum_{i=1}^{n} I((x_i, x_j) \in MST \wedge y_i \neq y_j)$
Ratio Intra/Extra Class Distance (N2)	Ratio between sum of intra-class nearest neighbor distances and sum of extra-class nearest neighbor distances.	$N2 = \frac{1}{1+\text{intra\_extra}}$
Local Set Cardinality (LSC)	Size of the local set formed by instances of the same class closer than the nearest enemy. Normalized by total same-class instances.	$LSC(x_i) = 1 - \frac{\lvert LS(x_i)\rvert}{\lvert\{x_j \mid y_j = y_i\}\rvert}$
Local Set Radius (LSR)	Normalized radius of the local homogeneous set around an instance. Larger radii indicate easier instances.	$LSR(x_i) = 1 - \min\{1, \frac{d(x_i, NE(x_i))}{\max(d(x_i, x_j) \mid y_j = y_i)}\}$
Distance/Density-based Measures		
Harmfulness (H)	Number of instances for which the current instance is the nearest enemy. Higher values indicate harder classification.	$H(x_i) = \frac{\lvert\{x_j \mid NE(x_j) = x_i\}\rvert}{\lvert\{x_j \mid y_j \neq y_i\}\rvert}$
Class Distribution-based Measures		
Majority Value (MV)	Measures the skewness of the class that an instance belongs to. For each instance, its MV is the ratio of the number of instances in the minority class to the total number of instances.	$MV(x_i) = 1 - \frac{\lvert\{x_j \mid y_j = y_i\}\rvert}{\max_{y_j \in Y} \lvert\{x_j \mid y_j = y\}\rvert}$
Class Balance (CB)	Measures the skewness of an instance's class by taking the proportion of instances that share the label of the instance in the dataset. The measure is adapted to be bounded in [0,1] interval.	$CB(x_i) = \left(1 - \frac{\lvert\{x_j \mid y_j = y_i\}\rvert}{n} + \frac{1}{C}\right) \times \frac{C}{C+1}$
Model-based Measures		
Disjunct Size (DS)	Builds an unpruned decision tree and verifies the relative size of the disjunct (leaf node) where the instance is placed. Larger disjunct sizes are expected for easier instances.	$DS(x_i) = 1 - \frac{\lvert\{x_j \mid x_j \in \text{Disjunct}(x_i)\}\rvert}{\max \lvert\text{Disjunct}(D)\rvert}$
Disjunct Class Percentage (DCP)	Complement of the percentage of same-class instances in the decision tree leaf containing the instance.	$DCP(x_i) = 1 - \frac{\lvert\{x_j \in \text{leaf}(x_i) \wedge y_j = y_i\}\rvert}{\lvert\text{leaf}(x_i)\rvert}$
Tree Depth Pruned (TD_P)	Normalized depth of the leaf node in a pruned decision tree. Deeper leaves suggest harder classification.	$TD\_P(x_i) = \frac{\text{depth}_{\text{leaf}}(x_i)}{\max \text{depth}}$
Tree Depth Unpruned (TD_U)	Normalized depth of the leaf node in an unpruned decision tree. Similar to TD_P but without pruning.	$TD\_U(x_i) = \frac{\text{depth}_{\text{leaf}}(x_i)}{\max \text{depth}}$
Class Likelihood (CL)	Posterior probability that the instance belongs to its true class according to a probabilistic classifier.	$CL(x_i) = P(y_i \mid x_i)$
Class Likelihood Difference (CLD)	Difference between the probability of the true class and the highest probability of any other class.	$CLD(x_i) = P(y_i \mid x_i) - \max_{y \neq y_i} P(y \mid x_i)$

Notes: $N_k(x)$ denotes the set of k nearest neighbors of instance x; $NE(x)$ denotes the nearest enemy (closest instance from a different class); $LS(x)$ denotes the local set of instance x; MST denotes the Minimum Spanning Tree; $I(\cdot)$ is the indicator function; n denotes the total number of instances in the dataset; C denotes the number of classes; Y denotes the set of all possible class labels; Disjunct(x) denotes the disjunct (decision tree leaf) containing instance x; $\max \lvert\text{Disjunct}(D)\rvert$ denotes the size of the largest disjunct in the dataset.